Practice workbook · How many patterns from n bits

Linking a number of bits to the number of distinct patterns/colours/levels, in both directions, including assigning unique codes with the minimum colour depth.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-08

1. Question 2 · part d.i · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

2 Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

(2)

(ii) Identify the number of characters that can be represented using standard ASCII.

(1)

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

(2)

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

(2)

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1. Question 2 · part d.i · 2 marks

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Worked-example retry · Attempt part d.i only.

(c) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	1	0	0
0	0	0	1	0	1	1	0

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

(2)

Colour	Binary pattern
Green	
Black	
White	
Red	
Blue	

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1. Question 2 · part d.i · 2 marks

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Worked-example retry · Attempt part d.i only.

(ii) Another image is 3579 pixels high and 6128 pixels wide.

The image is stored with a 32-bit colour depth.

The metadata for the image is 732 bytes.

Construct an expression to show how the file size, in **megabytes**, is calculated.

You do **not** need to do the calculation.

(4)

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(Total for Question 2 = 15 marks)

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Working and answer

2. Question 4 · part a.v · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.v only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

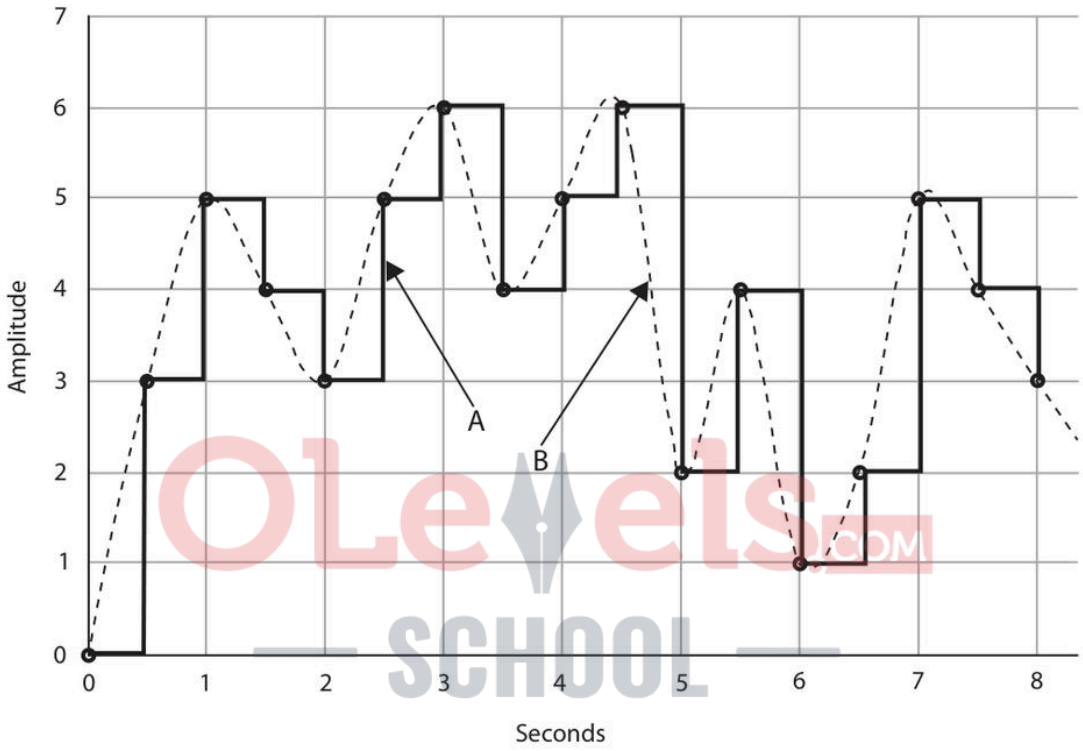


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

2. Question 4 · part a.v · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.v only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

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2. Question 4 · part a.v · 2 marks · Working and answer

Working and answer

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3. Question 1 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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3. Question 1 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

(f) Give the number of bits per character used by standard ASCII.

(1)

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

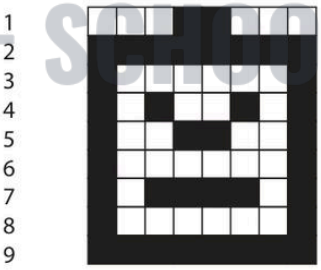


Figure 1

(i) State what is meant by the term **pixel**.

(1)

(ii) Construct an expression to show the number of pixels in the image.

(1)

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3. Question 1 · part c · 1 mark · Working and answer

Working and answer

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4. Question 1 · part g.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.iii only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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4. Question 1 · part g.iii · 1 mark

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Further practice · Attempt part g.iii only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

(f) Give the number of bits per character used by standard ASCII.

(1)

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

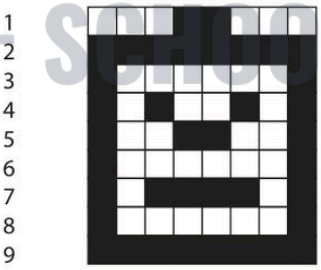


Figure 1

(i) State what is meant by the term **pixel**.

(1)

(ii) Construct an expression to show the number of pixels in the image.

(1)

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4. Question 1 · part g.iii · 1 mark

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Further practice · Attempt part g.iii only.

(iii) The image is changed so that any pixel can be one of 16 different colours.

State the minimum number of bits that would be needed to represent **one** pixel.

(1)

.....

(Total for Question 1 = 11 marks)

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5. Question 2 · part b · 1 mark

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Further practice · Attempt part b only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

5. Question 2 · part b · 1 mark · Working and answer

Working and answer

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6. Question 3 · part a.i · 1 mark

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Further practice · Attempt part a.i only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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6. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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